A Study of the Distribution of Popularity Using Wikipedia Page Views

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1 Introduction

Frequently, website hits on the Internet are associated with popularity and influence. Knowing this, the amount of times a person is researched on the online encyclopedia Wikipedia is indicative of the amount of influence a person has or has had over the course of history. I'd like to quantify how the probability distribution of historical influence changes relative to the industry a famous person works in.

For this project, we examined a dataset containing historical figures. The dataset obtained is from the Pantheon Project, through Kaggle, and describes these historical figures, giving the fields they worked in, their gender, their birthplace and birthdate, along with measures describing their occupation, field of industry, and general domain of influence. These are then combined with aggregate measures for the number of website hits each historical figure's Wikipedia entry received.

In this analysis, we'd like to examine the difference in distribution between the different industries. We want to see, for example, the probability distribution of the prominence of theoretical scientists, like mathematicians and statisticians, and compare it to the probability distribution for the prominence of team sports players. This would allow us to see to some degree how choice of field affect historical prominence. Furthermore, if one's sole aim was to become famous, we could use the posterior predictive distributions to see how many page views a person working in an industry can expect.

Our dataset has 11,341 historical figures. There are 88 distinct occupations, 27 industries these fields are grouped into, and 8 overall domains of influence. The minimum number of historical figures in an occupation is 1, and the maximum is 2530. Industries have more individuals in them with at least 10 historic figures in each industry, while the distribution of historic figures by domain is shown in Table 1. Numeric covariates, aggregate over the whole dataset, are shown in Table 2.

Before defining a sampling model or prior distributions, we made a few assumptions. Firstly, we assumed that the popularity of a historical figure is independent of the popularity of other historical figures.

2 Methods

For the natural logarithm of each historical figure's total number of website visits y_{ij}, we define the sampling model as

$$y_{ijk} \sim \text{Gamma}(\alpha_{ij}, \beta_{ij}),$$

where $i = 1, \ldots, N_D$ indexes domains, $j = 1, \ldots, M_i$ indexes industries within domain i, and $k = 1, \ldots, n_{M_i}$ indexes historical figures who worked in industry j. We select the Gamma likelihood because the data all together appears to be heavily right-skewed. We therefore suppose that the distribution for each industry j will also be right-skewed. Furthermore, we assume each historical figure's page's views is a random variable following the distribution of their industry. The likelihood will therefore be a product of Gamma densities. The likelihood is then

$$L(\alpha_1, \ldots, \alpha_{N_D}, \beta_1, \ldots, \beta_{N_D}; \boldsymbol{y}) = \prod_{i=1}^{N_D} \prod_{j=1}^{M_i} \prod_{k=1}^{n_{M_i}} \frac{\beta_{ij}^{\alpha_{ij}}}{\Gamma(\alpha_{ij}} y_{ijk}^{\alpha_{ij}-1} e^{-y_{ijk}\beta_{ij}} \tag{1}$$

Table 1: Categorical Variables

Variable	Count	Percentage
Domain		
Arts	2866	0.25
Business & Law	108	0.01
Exploration	102	0.01
Humanities	1329	0.12
Institutions	3456	0.30
Public Figure	358	0.03
Science & Technology	1366	0.12
Sports	1756	0.15
Sex		
Female	1495	0.13
Male	9846	0.87

Table 2: Numeric Variables

	Mean (SD)	Median (IQR)
Birth Year (CE)	997 (258)	1085 (125)
Latitude (degrees)	39.93 (18.69)	43.72 (13.02)
Longitude (degrees)	-7.04 (55.12)	4.48 (73.28)
Article Languages (Count)	41 (17)	35 (17)
Page Views (log)	14.36 (1.34)	14.29 (1.96)
Average Views	89414 (121461)	43844 (88808)
Historical Popularity Index	22.31 (3.36)	23.03 (4.16)

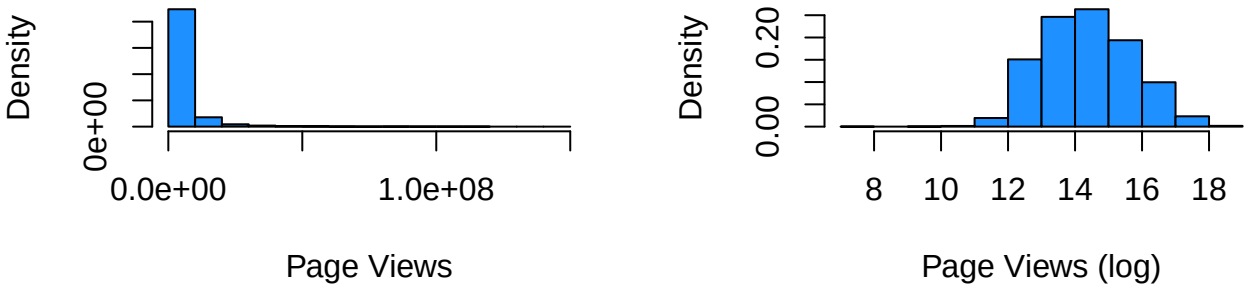


Figure 1: Histograms of Page Views

where $N_D = 8$ and $\sum_{i=1}^{8} M_i = 27$.

We define the response as the natural logarithm of page views to avoid issues with overflow. The number of page views is massive for many of our historical figures, such that the sum we calculate in obtaining the likelihood is assumed to be infinity. We took the natural logarithm because it is a one-to-one function that will overcome our overflow problem. Furthermore, from Figure 1 we see that the transformed data is still unimodal and appears to fit the constraints of our expected sampling model.

Our parameters α_{ij} and β_{ij} are positive by definition of the Gamma distribution. Because the Gamma distribution can fit many shapes on the positive real numbers, it seemed appropriate to assume α_{ij} and β_{ij} were distributed according to the Gamma distribution.

We don't have very much prior knowledge about the distribution of page views among historical figures, so we chose diffuse priors. For our prior distributions, we state that for each industry j within domain i, $\alpha_{ij} \sim \text{Gamma}(\lambda_i, \theta_i)$ and $\beta_{ij} \sim \text{Gamma}(\phi_i, \psi_i)$. For every other Gamma random variable or probability density function, we will use the rate parameterization of the Gamma distribution.

We introduce hierarchy to share information and account for the sparsity of data in some industries by setting hyper-parameters for each domain i. We assume that the parameters λ_i, θ_i, ϕ_i, and ψ_i are unique for each domain. Further, we set priors on these hyper-paramters, such that

$$\lambda_i \sim \text{Gamma}(a_i, b_i),\ \theta_i \sim \text{Gamma}(c_i, d_i),\ \phi_i \sim \text{Gamma}(f_i, g_i),\ \text{and}\ \psi_i \sim \text{Gamma}(p_i, r_i).$$

We incorporate our lack of knowledge about the distribution of page views into our choice of the hyper-parameters. We guessed, initially, that each set of hyper-parameters was distributed identically, such that for each i, $a_i = a$, $b_i = b$ et cetera for a group of numbers a, b, c, d, f, g, p, and r.

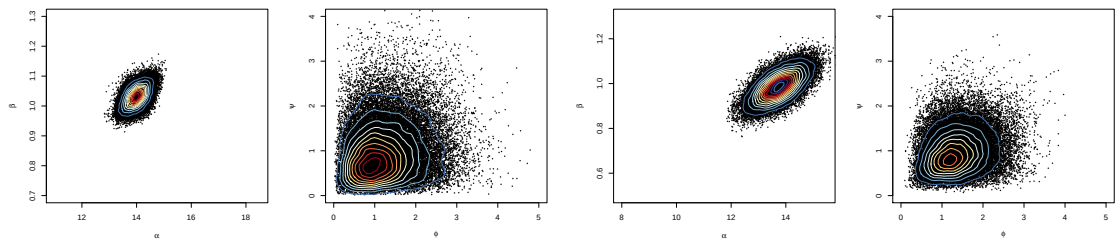
We chose the hyper-parameters using the average page views for all data. This may violate the Bayesian paradigm, but the priors are sufficiently diffuse as to convey our uncertainty. Because the distribution of page views is centered around 14, we select hyper-parameters such that the theoretic mean of the data is $\alpha\beta = 14$, because the mode of the overall data appears to be at about 14. A $\text{Gamma}(14, 1)$ distribution will do this. We chose the hyper-parameters accordingly, such that $a = 14$ and $b = c = d = f = g = p = r = 1$.

We performed a sensitivity analysis by alternating the amount of information in our hyper-priors. We consider three alternate sets of priors: a set of heavily informed priors, a set of noninformative hyper-priors, and a biased set of hyper-priors. The results of this analysis will inform us of how inaccurate priors can affect our analysis and are shown in Table 4. While our hyper-priors are justified, a sensitivity analysis will show how our choice to have diffuse hyper-priors affect the results of our analysis.

3 Results

Our results depend on a pair of assumptions. We assumed that the Gamma distribution would fit well to the data at hand and that the historical figures within our data are independent. However, other choices of likelihood are possible; entertained modeling the data using a Weibull sampling model. However, as the data shown in Figure 1 appear to be unimodal and have a positive support, our choice of using a Gamma sampling model is justified.

Our assumption of independence is harder to justify. While people certainly jump from one historical figure's page to another's when researching a topic, we assume that most of the visits are casual views, and not research delving into the history of a train of thought. We assume independence primarily because the amount of qualitative research needed to quantify the dependence between historical figures is prohibitive. For example, consider the rivalry between Magic Johnson and Larry Bird. Another statistics project could consist of creating a measure of dependence in popularity between such sets of rivals, if such a dependence were significant. We are not interested in the dependence between



(a) Joint Distribution of α and β for Math Industry

(b) Joint Distribution of ϕ and ψ for Science Domain

(c) Joint Distribution of α and β for History Industry

(d) Joint Distribution of ϕ and ψ for Social Science Domain

Figure 2: Posterior Distributions of Parameters and Hyper-parameters

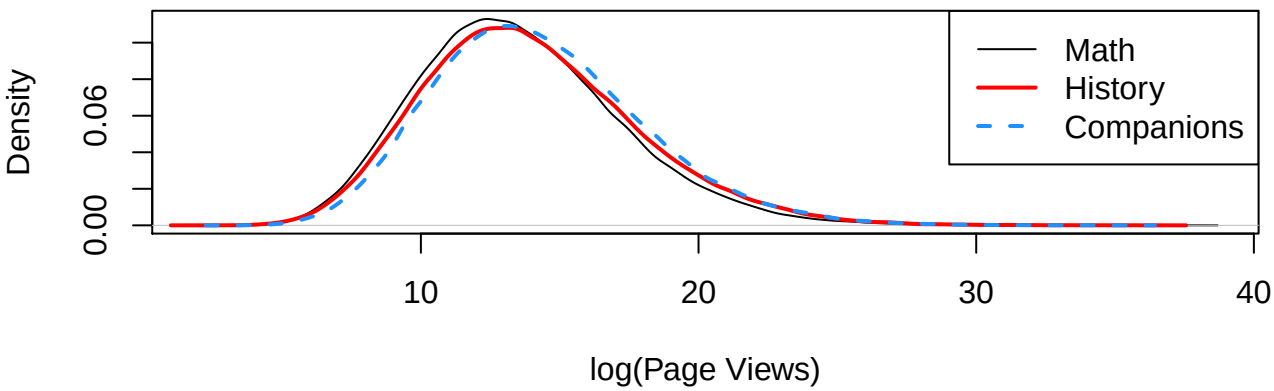


Figure 3: Posterior Predictive Distributions

historical figures, so we assume each historical figure's number of page views is independent of all other historical figure's number of page views.

Using Markov chain Monte Carlo (MCMC) methodology, we sampled from the posterior distributions for each of our parameters. For each industry, there are 2 parameters, and 4 more hyper-parameters for each of N_D domains, resulting in 86 total parameters to estimate. As such, we only included credible intervals for two industries in two different domains to show, to some degree, the contrast between industries in Table 3.

Figures 2a and 2b show the joint distribution of parameters α and β for the Philosophy industry and hyper-parameters λ and θ for the History domain. We can contrast these figures with the joint distribution of the same parameters and hyper-parameters for the Outlaws industry and the Public Figure domain, found in Figures 2c and 2d.

The differences between these distributions are not readily apparent. We are, however, a bit more interested in the posterior predictive distribution for drawing inference on actual historical figures and their industries. The posterior predictive distributions for Math, History, and Outlaws industries are shown in Figures 3.

There doesn't appear to be any significant difference between the posterior predictive distributions. While there is some difference, quite a bit of overlap exists between the two industries, such that there is a relatively similar probability the log page views is less than 14 for each of the two industries.

There is a significant difference between the posterior theoretical means for many of our industries. We used our draws from the posterior distribution to create credible intervals for the posterior mean α/β. The 95% predictive intervals do not overlap for some of our industries, indicating that although the posterior predictive distribution is similar for many of our industries, our estimate of the posterior mean is sometimes significantly different, as shown in Figure 4.

Table 3: Bayesian Credible Intervals on Model Parameters for Select Industries

Quantile	Math		History		Companions	
	α	β	α	β	α	β
0.5%	13.990	1.031	13.738	0.986	14.672	1.030
0.025%	13.424	0.973	12.638	0.880	13.918	0.957
0.975%	14.563	1.092	14.860	1.098	15.435	1.106

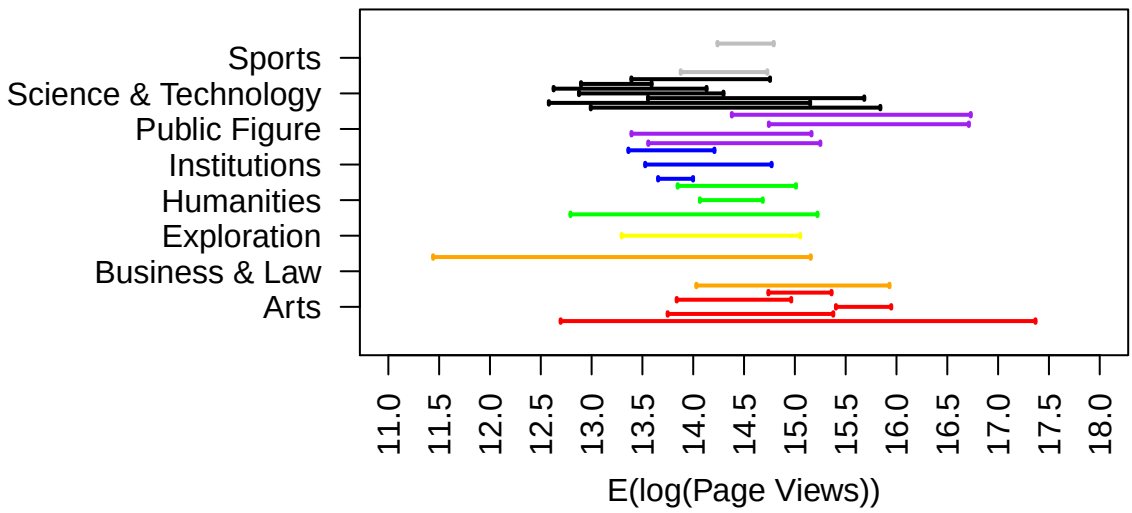


Figure 4: Bayesian Predictive Intervals on Theoretical Means for Industries within Domains

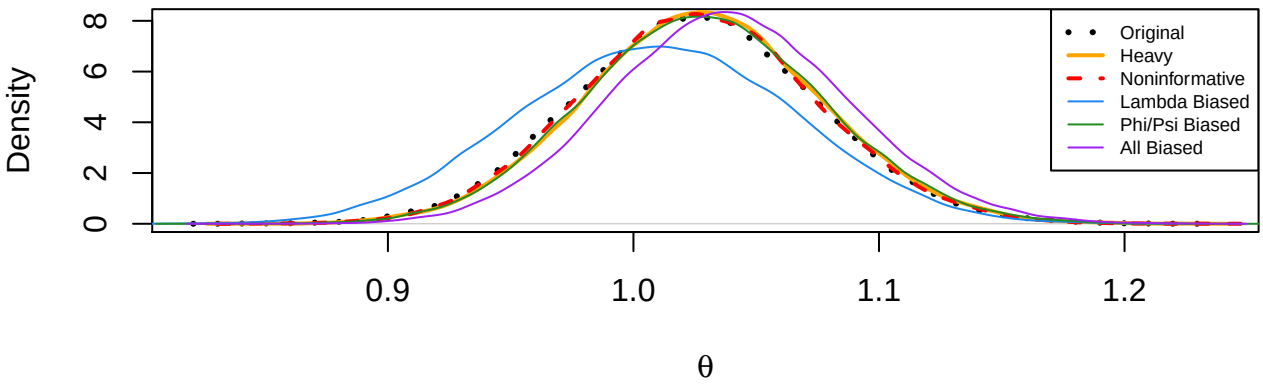


Figure 5: Posterior Distributions for $\lambda_2 3$ From 4 Models in Sensitivity Analysis

We ran a number of other models with the sets of hyper-priors described in the Methods section. The results of this sensitivity analysis can be seen in Table 4. We see that all models fit in a similar fashion, such that the model diagnostics were very similar. Furthermore, in examining the posterior distributions, we found that the conclusions reached were very similar. The marginal posterior distribution for λ_{11} for each of our four models is shown in Figure 5.

From Table 4, the average Gelman diagnostic can be seen. Note that for our chosen analysis, $\hat{R}$ is close to 1, indicating that convergence was reached with a high level of accuracy. This was also true throughout our sensitivity analysis. A mixing plot of our multiple chains for the hyper-parameters involved in modeling the distribution of Activism individuals' page views is also shown in Figure 6 as an example.

We also carried out a frequentist analysis of our data. We defined the likelihood in the same way as in Equation

Table 4: Sensitivity Analysis of Alternate Priors. $\hat{R}$ is the average Gelman Diagnostic.

	Hyperparameters								Model Diagnostics		
Description	a	b	c	d	f	g	p	r	AIC	DIC	$\hat{R}$
Original Analysis	14	1	1	1	1	1	1	1	38222	51919	1.034
Heavily Informed	1381551	100000	1000	1000	1000	1000	1000	1000	38222	51919	1.282
Uninformative	0	0	0	0	0	0	0	0	38222	51930	1.282
λ, θ Biased	1000	14000	1	1	1	1	1	1	38222	51963	1.282
ϕ, ψ Biased	14	1	1	1	10	1000	10	1000	38222	51911	1.282
All Biased	1000	14000	10	1000	10	1000	10	1000	38222	51928	1.282
Frequentist	-	-	-	-	-	-	-	-	38222	-	-

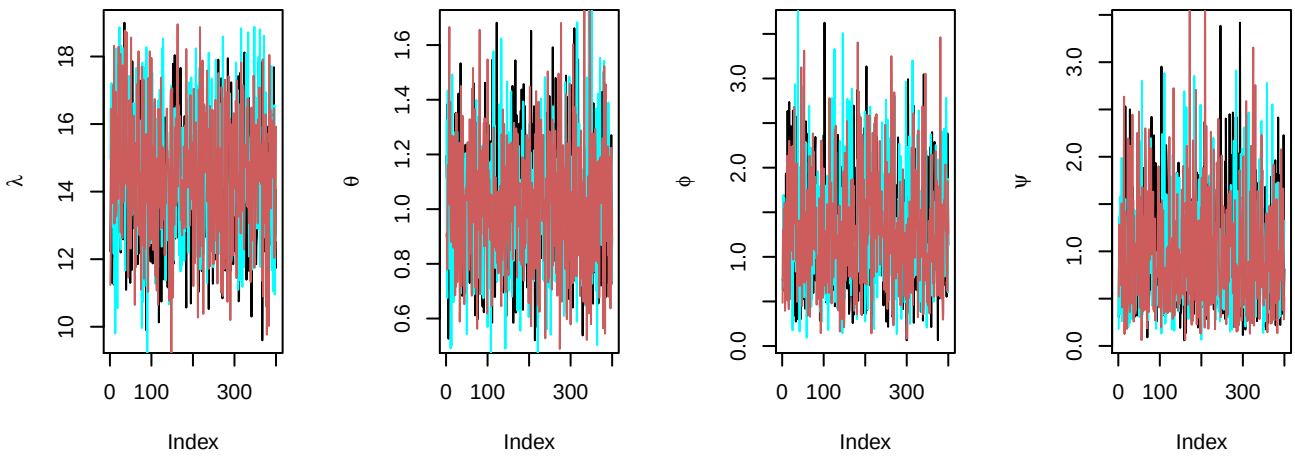


Figure 6: Mixing Plots with Multiple Chains for Certain Model Parameters for the Activism Industry

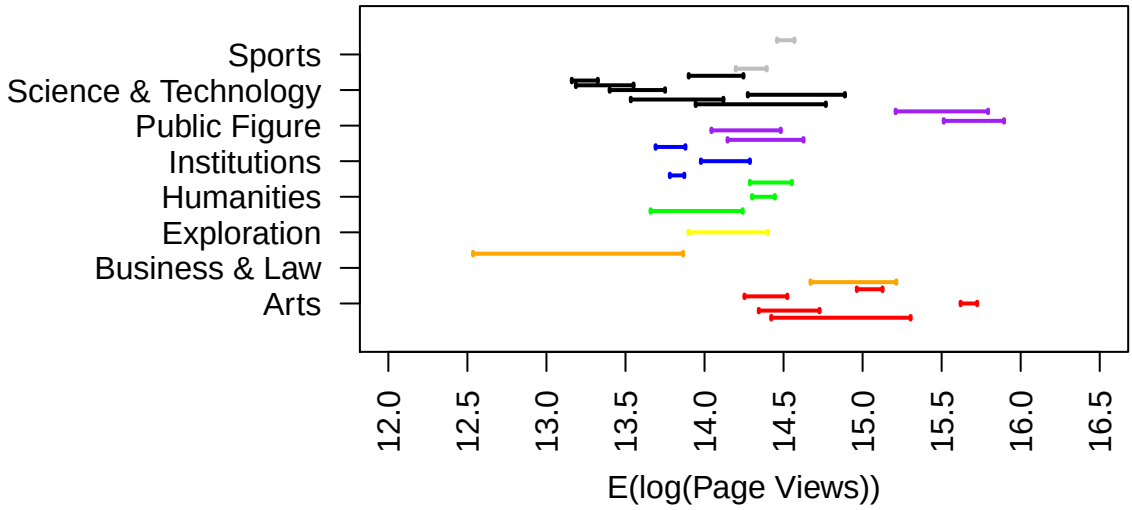


Figure 7: Bootstrap 95% Confidence Intervals for the Theoretical Mean for Each Industry

1 and found the Maximum Likelihood Estimates (MLE). However, the MLE for each of our parameters could not be found in closed form, relative to our data, so we employed the R function optim to find the parameter combinations that maximized the likelihood. We then used the bootstrap to get 95% confidence intervals on each of our parameters. Similar to Table 3, Table 5 contains our estimates for the parameters within two industries and two domains for our frequentist analysis, along with 95% confidence intervals. Figure 7 also shows how the theoretical mean for each industry's log page views vary relative to one another. We can also see from Table 4 that the AIC for our frequentist analysis was quite a bit higher than our Bayesian analysis.

Our model fitting used a Markov chain Monte Carlo methodology, such that we will fit each parameter in serial a large number of times. Roughly half of the parameters can be sampled using Gibbs sampling, with the remaining parameters being sampled using a more general Metropolis-Hastings algorithm. We will sample three chains of $50,000$ samples from each posterior distribution, and discard the first $10,000$ samples to account for burn-in. Each chain takes roughly 3 minutes to run. Starting values for the analysis were chosen by drawing from a uniform distribution with bounds one and ten.

Table 5: Frequentist 95% Bootstrap Confidence Intervals on Model Parameters for Select Industries

Quantile	Math		History		Companions	
	α	β	α	β	α	β
0.5%	129.52	9.46	132.78	9.55	123.84	8.70
0.025%	157.93	11.65	196.30	14.01	154.90	10.85
0.975%	203.51	15.15	325.87	23.23	200.94	14.18

DIC, WAIC, and LPML calculations were costly in terms of computation. Calculating DIC with only a tenth of our posterior retained after thinning took about 30 seconds for each analysis. WAIC was much more costly; it took 2 minutes to calculate the WAIC for 100 of our posterior draws. LPML was also quite costly. Using 100 posterior draws, it took 40 seconds to calculate LPML for each model. Overall, it seemed that only the DIC was worth calculating. The other diagnostics didn't add very much information and took a lot of time computationally.

We therefore conclude that the analysis is valid, because its convergence diagnostics indicated a good fit, our model diagnostics indicate a good fit, and our assumptions in this analysis seem justified. Furthermore, we can see from the differences in AIC contained in Table 4 that the Bayesian analysis outperforms the Frequentist analysis with regards to model fit. The conclusions we reached by fitting the model are valid, such that we conclude there is no significant difference between the number of page views within industries, although there is a significant difference between the average number of page views between industries.

4 Conclusions

We see from our analysis that although the average number of views for each industry was different, the distributions of the page views by industry were all fundamentally the same, with large amounts of overlap between the posterior predictive distributions. There is no silver bullet to getting page views once one is famous, all industries seem to incorporate similar amounts of variability with slightly different means.

Further steps could include an analysis that applies a different sampling model. Due to the skewness of the data, I suspect a Weibull sampling model may fit better, but the difference in parameterization between R and the parameterization I use was somewhat prohibitive. Furthermore, an analysis that attempts to incorporate dependence between Wikipedia pages by examining hyperlinks between pages or using another dataset to connect historical figures.